INTERNSHIP REPORT

Submitted

For the Partial Fulfilment of the Requirement for the Degree of Bachelor of _____
with _____ (Major and Minor Subjects) under **Magadh University**

Submitted to

The Department of _____, _____ **(College Name)**

[Paste College Logo here]

Submitted by

Name : _____

Semester : _____

Programme FYUGP : _____

University's Registration No. : _____

College Roll No. : _____

Year : _____

RK NEXORA
CAMPUS TO CAREER
A Unit of RKNexora Private Limited.

INTERNSHIP REPORT

Topic - Data Analytics

Submitted to the Head of Department, _____ (College Name), in partial fulfilment of the requirements for the Degree of _____, Magadh University

Name of Student : _____

Programme : _____

University Registration No. : _____

College Roll No. : _____

Internship Registration No. : _____

Name of the IPO / Agency : RKNEXORA PRIVATE LIMITED

Internship Period : _____ to _____

Semester : _____

Year / Session : _____



_____ _____

Signature of Student **Seal & Signature of IPO**

ACKNOWLEDGEMENT

I express my sincere gratitude to RK NEXORA, RKNEXORA PRIVATE LIMITED, _____ (College Name), my faculty mentors and the internship supervisor, _____, for their guidance and support. The structured activities, learning resources and regular feedback helped me complete this internship in Data Analytics successfully.

I am thankful to the Head of the Department and the college mentor for their encouragement and for guiding me in preparing this report. I am also grateful to the management of RKNEXORA PRIVATE LIMITED for providing a professional learning environment and the opportunity to learn and apply my knowledge in a practical setting. The concept sessions, hands-on tasks with sample datasets and demo files, and the supervisor's patient and detailed feedback helped me understand how data is cleaned, analysed and turned into decisions behind the scenes.

Finally, I thank my family and peers for their constant encouragement throughout the programme.

_____

Name: _____

University Registration No. - _____

INTERNSHIP COMPLETION CERTIFICATE

The Internship Completion Certificate issued by RKNEXORA PRIVATE LIMITED for the internship in Data Analytics is to be attached on this page.

[Attach the original / printed copy of the QR-verified Internship Completion Certificate here]

Certificate No. _____

CONTENTS

Acknowledgement....3
Internship Completion Certificate....4
Preface....6
Executive Summary....7
About the Agency....8
Plan and Execution....9
Learning Outcome....22
Conclusion....24
Attendance Sheet....26

PREFACE

Internships are an important part of the undergraduate programme, as they allow students to see how classroom learning is applied in real organisations. The Four-Year Undergraduate Programme (FYUGP), introduced in line with the National Education Policy 2020, gives special importance to experiential learning, skill development and employability. This report has been prepared as part of the internship requirement of Magadh University and the Higher Education guidelines applicable in Bihar.

The report describes the internship undertaken in Data Analytics at RKNEXORA PRIVATE LIMITED, the plan followed, the work carried out, and the learning gained. Every shop, school, hospital, bank and office today records its work in numbers: daily sales, fees collected, patients treated, loans given, attendance and enquiries. When these records are only stored, they help no one; when they are cleaned, summarised and explained, they show which product sells most, which month is weak, which customers return and where money is being lost. Data analytics is the set of methods that makes this possible: spreadsheets, data cleaning, database queries, statistics, charts, dashboards and clear reporting. A small shopkeeper in Gaya and a national company can both use it to take better decisions. This internship offered a first-hand introduction to how these methods work together.

The report has been arranged in the order suggested by the University, beginning with the cover page and acknowledgement and ending with the conclusion and attendance sheet. It is hoped that the report gives a clear and honest account of the internship experience. All examples in it use dummy data and fictitious businesses; no real customer or business information was handled.

EXECUTIVE SUMMARY

This report documents the internship completed in Data Analytics at RKNEXORA PRIVATE LIMITED (RK NEXORA - 'Campus to Career'). The programme required a minimum of 120 hours, completed over about four to five weeks (28 working days) with daily sessions of four to five hours, and was supported by an attendance log, supervisor guidance and review.

Background. Data analytics means collecting, cleaning, examining and interpreting data so that decisions are based on evidence and not on guesswork. Organisations of every size now produce data every day, but the value lies in turning that data into clear answers: what happened, why it happened, what is likely to happen next and what should be done. A good analyst therefore combines accuracy with curiosity and communication. The work covers understanding the business question, summarising data in Excel, cleaning it, asking questions of a database with SQL, using statistics to measure patterns, choosing honest charts, building dashboards in Power BI or Tableau, exploring datasets and reporting findings in plain language. For a student from any discipline, the internship shows how mathematics, logic, business sense, technology and writing come together in a real analytics job.

Objectives. The internship aimed to (i) build a structured understanding of data analytics, its types, process and tools, (ii) learn Excel for data analysis, including formulas, filters, pivot tables and charts, (iii) learn data cleaning and SQL queries, (iv) learn statistics for analytics and data visualization, (v) learn Power BI / Tableau basics, exploratory data analysis, dashboard creation and insight reporting, and (vi) develop communication, documentation and professional discipline.

Work carried out. The student first studied the overview of data analytics, then worked through Excel for data analysis, data cleaning, SQL queries, statistics for analytics, data visualization, Power BI / Tableau basics, exploratory data analysis, dashboard creation and insight reporting. Each topic had a concept session followed by a hands-on task using free tools, sample datasets and dummy data. The later part of the programme was devoted to a case study, a mock practical exercise, supervisor review and preparation of this report.

Project. The work was tied together in a project titled *Business Data Analysis & Dashboard Project*, in which a sample business dataset was cleaned, analysed with Excel and SQL, presented in a dashboard and summarised in an insight report.

Method. A blended method was followed - learn, practise, document and review - supported by a daily attendance log with day-wise remarks and regular feedback from the supervisor.

Learning gained. The student gained a clear understanding of how data is prepared, analysed and presented, practical familiarity with Excel formulas and pivot tables, data cleaning logs, SQL queries with grouping and joins, descriptive statistics and correlation, chart selection, Power BI / Tableau measures and visuals, KPI dashboards and insight reports, awareness of data protection and ethical rules, and improved communication, analytical thinking and professional discipline.

Key observations. Analysis is only as good as the data behind it, so cleaning comes first; every result should be checked against an independent total; the chart must fit the question; a dashboard must have a clear purpose and defined KPIs; and reports should lead to decisions and not merely list numbers.

Conclusion. The internship met its objectives and contributed to the student's academic understanding, employability and confidence. The report follows the University's suggestive format: cover page, acknowledgement, certificate, contents, preface, executive summary, about the agency, plan and execution, learning outcome, conclusion and attendance sheet.

ABOUT THE AGENCY

RK NEXORA, a unit of RKNEXORA PRIVATE LIMITED, provides structured internship and skill-development programmes under the theme 'Campus to Career'. The organization focuses on academic-industry alignment, practical exposure, assessment, documentation and employability-oriented learning.

Particulars	Details
Name of Agency	RKNEXORA PRIVATE LIMITED
Programme / Unit	RK NEXORA - Campus to Career
Registered Office	C/O Mukesh Kumar, Khiri, PS Atri, Jethiyan, Atri, Gaya - 805236, Bihar, India
CIN	U85499BR2026PTC086757
Email	rknexorapvtltd@gmail.com
Website	www.rknexora.org
Internship Topic	Data Analytics

1. Vision

To support students in developing practical, professional and industry-relevant capabilities.

2. Mission

To deliver structured learning experiences that connect academic knowledge with workplace expectations.

3. Activities of the Agency

1. Structured internship programmes aligned with university and higher-education requirements.
2. Skill-development learning modules, practical activities and assignments.
3. Attendance tracking, supervision, assessment and documentation.
4. Employability-oriented guidance connecting campus learning to career readiness.

4. Programme Structure

The internship programme is organised around a fixed number of learning hours, completed over a defined period. Each intern is registered on the internship portal with a unique registration number, is assigned a supervisor, follows a day-wise plan and maintains an attendance log. Learning is organised into three phases: **Orientation** (Days 1-2), **Learning and Practice** (Days 3-22, ten sub-topics of two days each: one concept day and one hands-on day) and **Project and Review** (Days 23-28), followed by a concluding review.

5. Role of the Supervisor

The internship supervisor assigned by the agency guided the student through the programme, explained concepts, assigned practice tasks, reviewed the daily log and gave feedback. The supervisor also verified attendance and reviewed the student's work at the end of the programme.

PLAN AND EXECUTION

1. Objectives

The internship was planned around the following objectives, each linked to a part of the programme:

1. Understand Data Analytics - meaning, types of analytics, the analytics process and common tools (Week 1).
2. Learn Excel for data analysis - formulas, sorting, filters, pivot tables and charts (Week 1).
3. Learn data cleaning and SQL queries - removing duplicates, fixing errors and handling missing values; SELECT, WHERE, GROUP BY, JOIN and basic queries (Week 2).
4. Learn statistics for analytics - mean, median, variance, correlation and distributions (Week 2).
5. Learn data visualization, Power BI / Tableau basics and exploratory data analysis - choosing charts, connecting data, building visuals and reports, summarising and finding patterns in datasets (Week 3).
6. Learn dashboard creation and insight reporting - an interactive dashboard with KPIs and filters, and turning analysis into findings and recommendations (Week 4).
7. Develop professional discipline - communication, honesty, documentation and report writing (throughout).

2. Methodology

The programme followed a blended method made up of four connected steps, repeated for each topic:

Learn. Each topic began with a concept session in which the supervisor explained the purpose of the technique or tool, its main terms and how it connects with the other parts of an analysis. Notes were prepared in the student's own words.

Practise. The concepts were then applied in guided hands-on tasks - writing formulas and building pivot tables on a sample sales sheet, cleaning a messy dummy file, writing SQL queries on a sample database, calculating statistics in Excel, choosing and drawing charts, connecting data to Power BI or Tableau, exploring a dataset, designing a dashboard and writing an insight report. Only fictitious businesses, sample datasets and dummy data were used, so that no real customer or business information was handled.

Document. Every day's work was recorded in the internship attendance log with a short remark, and workbooks, queries, dashboards and drafts were saved so that the work could be reviewed.

Review. The supervisor reviewed the work, pointed out errors and suggested improvements. Review days, a case study and a mock practical exercise helped to consolidate the learning.

3. Understanding Data Analytics - Concepts Studied

3.1 Data analytics fundamentals

Data analytics is the process of collecting, cleaning, examining and interpreting data in order to find useful patterns and support better decisions. Every organisation, from a small shop in Gaya to a large bank, produces data every day: sales entries, customer enquiries, attendance records, website visits and payments. On its own this data is only a collection of numbers and words; analytics turns it into information that a manager can act upon. Data may be structured (organised in rows and columns, such as a sales table or a spreadsheet), semi-structured (such as JSON or XML files) or unstructured (such as emails, photographs, audio and free text). It may also be quantitative (numbers that can be measured) or qualitative (descriptions and opinions). A single record is one row of a dataset, and each column is a field or variable.

Analytics is commonly divided into four types. **Descriptive analytics** answers "what happened?" through totals, averages and charts. **Diagnostic analytics** answers "why did it happen?" by comparing groups, drilling down and looking for causes. **Predictive analytics** answers "what is likely to happen?" using trends, statistics and models. **Prescriptive analytics** answers "what should we do?" by recommending actions. At the internship level the work was mainly descriptive and diagnostic, with an introduction to simple prediction through trends.

A typical analytics project follows a sequence of steps. First, the business question is stated clearly, for example "Which products should the shop stock more of next month?" Second, the data needed is identified and collected

from spreadsheets, databases or forms. Third, the data is cleaned. Fourth, it is explored and analysed with formulas, queries and statistics. Fifth, the results are shown through charts and dashboards. Sixth, findings are explained in plain language with recommendations. Finally, the decision is taken and the results are followed up. The common tools are MS Excel and Google Sheets for everyday analysis, SQL for working with databases, Power BI and Tableau for visualisation, and programming languages such as Python or R for advanced work. The people who use analytics include data analysts, business analysts, MIS and reporting executives, and managers, and a good analyst combines curiosity, accuracy, business understanding and the ability to communicate.

3.2 Excel for data analysis

MS Excel is the most widely used tool for day-to-day data analysis, because it is easy to learn and can handle thousands of rows. Good practice begins with a tidy layout: one header row, one record per row, no blank rows or merged cells inside the data, and the range converted into an Excel Table so that formulas and charts extend automatically when rows are added. Basic formulas include SUM, AVERAGE, MIN, MAX, COUNT and COUNTA. Conditional formulas include IF, COUNTIF, COUNTIFS, SUMIF and SUMIFS, which count or add only the rows that meet a condition. Lookup formulas such as VLOOKUP, INDEX with MATCH, and the newer XLOOKUP bring information from another table using a common key such as a product code. Text functions such as TRIM, LEFT, RIGHT, UPPER and CONCAT, and date functions such as TODAY, MONTH and YEAR, help in preparing data. Absolute references (for example B2) keep a cell fixed while a formula is copied, and a formula should never be replaced by a typed number if the result is meant to update.

Sorting arranges records by one or more columns, and filters show only the rows that match chosen conditions, while conditional formatting highlights values such as the lowest sales or overdue payments. Data validation restricts entries to allowed values and so prevents errors at the source. A **pivot table** is the most powerful summary tool: by dragging fields into rows, columns, values and filters, a long list of transactions can be summarised by category, month or city within seconds, with sum, count, average or percentage of total. Pivot tables must be refreshed after the source data changes, and slicers give point-and-click filters. Charts such as column, bar, line and pie charts, and PivotCharts linked to a pivot table, present the summary visually. Totals should always be checked against an independent figure, for example the grand total of the raw data, before a result is reported.

3.3 Data cleaning

Real data is almost never perfect. Data cleaning (also called data preparation or data wrangling) is the work of finding and correcting problems before analysis, and analysts often spend more time on it than on any other step, because wrong data leads to wrong decisions however good the charts look. Common problems are duplicate records, spelling variations of the same value (for example "Gaya", "gaya " and "GAYA"), extra spaces, inconsistent date and number formats, numbers stored as text, missing values, impossible values (such as a negative quantity or an age of 250), and outliers that may be errors or genuine exceptional cases.

A sound cleaning routine starts by keeping an untouched copy of the raw data, then profiling it: counting records, checking each column for blanks, unusual values and data types. Duplicates are removed after deciding which columns define a duplicate. Text is standardised with TRIM, PROPER or UPPER and with Find and Replace or lookup lists. Dates and numbers are converted to proper types. Missing values are handled according to the situation: the record may be deleted if little information is lost, the value may be filled with a sensible substitute such as the median or the most common value, a label such as "Unknown" may be used for categories, or the value may be left blank and reported as missing - and whichever choice is made must be written down. Outliers are checked with the supervisor or source before they are removed. Excel features such as Remove Duplicates, Text to Columns, Flash Fill, filters and Power Query make these tasks repeatable. The final step is validation - comparing record counts and totals before and after cleaning - and a **cleaning log** that records every change. Data quality is judged on accuracy, completeness, consistency, validity, uniqueness and timeliness.

3.4 SQL queries

SQL (Structured Query Language) is the standard language for asking questions of a relational database, in which data is stored in tables made of rows (records) and columns (fields). Each table has a **primary key**, a column whose value is unique for every row (such as order_id), and tables are connected through **foreign keys**,

columns that hold the primary key of another table (such as customer_id in a sales table). Software such as MySQL, PostgreSQL, SQL Server, Oracle and SQLite all use SQL with small differences in syntax, so the commands learned apply widely.

The basic query uses SELECT to choose columns and FROM to name the table. WHERE filters rows using comparison operators and AND, OR, NOT, IN, BETWEEN and LIKE; missing values are tested with IS NULL and not with an equals sign. ORDER BY sorts the result in ascending (ASC) or descending (DESC) order, and DISTINCT removes repeated values. Aggregate functions - COUNT, SUM, AVG, MIN and MAX - calculate one value from many rows, and GROUP BY applies them to each group separately, for example total sales for each category. HAVING filters the groups after aggregation, whereas WHERE filters rows before it. A **JOIN** combines rows from two tables on a matching column: INNER JOIN keeps only matching rows, LEFT JOIN keeps all rows of the left table and fills the missing matches with NULL, and aliases (such as s for sales) keep queries short. Although a query is written starting with SELECT, the database processes it in the order FROM, WHERE, GROUP BY, HAVING, SELECT and ORDER BY, which explains why a column alias cannot be used in WHERE. Good practice includes naming columns instead of using SELECT *, testing queries on small samples, checking results against known totals and using a read-only practice database so that live data is never changed. Commands that change data, such as UPDATE and DELETE, must be used with great care and always with a WHERE condition.

3.5 Statistics for analytics

Statistics gives analytics its method for summarising data and judging how far a pattern can be trusted. Measures of central tendency describe the typical value: the **mean** is the sum divided by the number of values, the **median** is the middle value when the data is sorted, and the **mode** is the most frequent value. The mean is affected by very large or very small values, while the median is not, so for skewed data such as incomes or order values the median is often the better summary. Measures of spread show how much values differ: the range (maximum minus minimum), the **variance** (the average squared distance from the mean) and the **standard deviation** (the square root of the variance, in the same unit as the data). Excel provides separate functions for a sample (VAR.S and STDEV.S, which divide by n minus 1) and for a whole population (VAR.P and STDEV.P). Percentiles and quartiles divide sorted data into parts, and the interquartile range (IQR) is used in a common rule that treats values below Q1 minus 1.5 times IQR or above Q3 plus 1.5 times IQR as possible outliers.

Correlation measures how two numerical variables move together, on a scale from -1 (perfect opposite movement) through 0 (no linear relationship) to +1 (perfect same movement), and is calculated in Excel with the CORREL function. A strong correlation does not prove that one thing causes the other; hidden factors such as a festival season may affect both. A **distribution** shows how values are spread: a normal distribution is bell-shaped and symmetrical, while a skewed distribution has a long tail on one side. Histograms and box plots show the shape. Further ideas studied at an introductory level were sampling (studying a part to learn about the whole, which must be chosen fairly to avoid bias), basic probability, percentages and percentage change, and the meaning of a trend, seasonality and a moving average. Averages and percentages must always be read together with the number of records behind them, because a high percentage from a very small group can be misleading.

3.6 Data visualization

Data visualization is the presentation of data as charts, maps and tables so that patterns that are hidden in rows of numbers become clear at a glance. The first rule is to start from the question and the audience, and to choose the chart that fits the question. A bar or column chart compares amounts across categories; a line chart shows change over time; a pie or donut chart shows parts of a whole and should be used only for a few categories; a scatter plot shows the relationship between two numbers; a histogram shows how values are distributed; a box plot compares spread between groups; a heat map or matrix shows two-way comparisons; and a table is best when exact values matter. Three-dimensional effects, too many colours, and charts with more than a few series are usually avoided.

Good charts have a clear title that states the message, labelled axes with units, a sensible order (for example bars sorted from largest to smallest), a baseline that starts at zero for bar charts, restrained and consistent colour used to highlight what matters, and no unnecessary decoration. Charts can mislead, deliberately or by accident, through truncated axes, selective time periods or confusing scales, so honest design is part of professional ethics. Colour choices should also work for people with colour-vision difficulty and when printed in black and white. The aim of data storytelling is to guide the audience from a question, through the evidence, to a conclusion.

3.7 Power BI and Tableau basics

Power BI and Tableau are business intelligence (BI) tools that connect to data, build interactive visuals and share reports. In **Power BI Desktop** the usual flow is: get data from Excel, CSV files or databases; transform it in Power Query, where cleaning steps are recorded and can be repeated when the data is refreshed; build a data model by creating relationships between tables (for example Sales and Customers linked by customer ID, in a one-to-many relationship); create measures using DAX (Data Analysis Expressions), for example Total Sales = SUM(Sales[Amount]); and place visuals such as cards, bar charts, line charts, maps and tables on report pages, with slicers for filtering. Reports are published to the Power BI service to be shared, with scheduled refresh. In **Tableau** the data is connected on the data source page; fields are classed as dimensions (categories such as city) and measures (numbers such as sales); visuals are built on worksheets by dragging fields onto rows, columns, marks and filters; calculated fields create new values; and worksheets are combined into a dashboard. Tableau Public is a free version, but everything saved to it is publicly visible, so only dummy or openly published data should be used with it.

Both tools reward a clean, well-structured data model: a fact table of transactions linked to small dimension tables such as customers, products and dates is easier to analyse than one very wide sheet. Tools differ in price, features and licensing, and new versions appear often, so the choice depends on the organisation. The skills learned - connecting, shaping, modelling, visualising and sharing data - are common to both.

3.8 Exploratory data analysis

Exploratory data analysis (EDA), a term made popular by the statistician John Tukey, is the first careful look at a dataset to understand its structure, quality and patterns before any formal conclusions are drawn. The steps are: find out the size of the data, the meaning and type of each column and the period covered; check missing values, duplicates and impossible entries; summarise each numeric column with count, mean, median, minimum, maximum and standard deviation; study each category column with counts and percentages; draw histograms and box plots to see distributions and outliers; compare groups (for example sales by city, category or weekday); look at trends over time; examine relationships between pairs of numbers with scatter plots and correlation; and finally write down the patterns found, the surprises and the questions that need further study.

EDA is guided by curiosity but must be disciplined. Findings should be checked from more than one angle, because patterns can appear by chance, especially in small samples, and a correlation is not proof of cause. Every assumption and every filter applied should be noted so that another person can repeat the work and reach the same result.

3.9 Dashboard creation

A dashboard is a single screen that shows the most important measures of a business in a way that can be read at a glance and explored through filters. Good dashboards begin with the audience and purpose: a shop owner who wants a daily view needs different information from a manager who reviews a quarter. The key performance indicators (KPIs) are chosen carefully and defined in writing - for example total sales, number of orders, average order value, profit margin and growth over the previous period - and each is compared with a target or an earlier period so that a number has meaning. The layout follows reading order: the most important KPI cards at the top, trends and comparisons in the middle, and details lower down. Interactive features include slicers or filters for period, city and category, drill-down from summary to detail, and tooltips that give extra information on hover.

Good practice includes limiting a page to a handful of visuals, using consistent colours, fonts and number formats, aligning items in a grid, avoiding clutter, testing performance with realistic data volumes, adding a short note on data source, date of refresh and definitions, and checking every figure against the source before sharing. A dashboard that nobody understands or trusts has failed, however attractive it looks, so it should be tested with the people who will use it, and maintained when the business or the data changes.

3.10 Insight reporting

An insight is a finding that is supported by evidence and that suggests an action; it is more than a description of numbers. "Sales were Rs. 4,000 on Monday" is only a fact, while "Saturday sales are about one-third higher than the weekly average, so more stock and staff are needed on Saturdays" is an insight. Reporting follows a simple chain: finding, evidence, meaning and recommendation. A good report opens with an executive summary giving the most important results in a few lines, states the business question and the data used, presents three to five key

findings each supported by one clear chart or number, explains the reasons or likely causes, gives specific and prioritised recommendations with the person responsible, and lists the limits of the data and the next steps.

The language should be simple and free of unexplained technical terms, because the reader is usually a decision-maker and not an analyst. Honesty is essential: results must not be exaggerated, unfavourable findings must be shown, assumptions must be stated, and uncertainty must be admitted, for example when the sample is small or when the data covers only a short period. The same method should be used in each regular report so that periods can be compared.

3.11 Rules, ethics and limitations

Data analysts handle information that may be personal, confidential or commercially valuable, and so must follow the law and professional ethics. The **Digital Personal Data Protection Act, 2023** requires that personal data, such as names, phone numbers and purchase records of identifiable people, be used lawfully, for a stated purpose and with consent, and that only the data needed be collected. The **Information Technology Act, 2000** governs electronic records, unauthorised access and cyber offences. The **Copyright Act, 1957** protects datasets, reports and software, and licences and terms of use of datasets and tools must be respected. Organisations also have their own confidentiality and data-security policies and non-disclosure agreements.

Good practice includes using only the data one is authorised to use, removing or masking names, phone numbers and other personal identifiers before sharing (anonymisation), keeping files in secure places with limited access, never sending sensitive data through personal messaging apps, keeping a copy of raw data unchanged, documenting all steps, giving credit for data sources, and being open about the limits of an analysis. Analysts must also watch for bias - for example data collected only from one group of customers may not represent everyone - and must not present a chart or a number in a way that is meant to mislead. The tools, functions, menu names and legal provisions mentioned in this report change over time, so they must be verified against official documentation, and the figures in the examples are dummy values for learning only.

4. Week-wise Plan and Execution

Week	Days	Activities
Week 1	Days 1-6	Orientation and foundations: company induction, introduction to data analytics, data analytics overview, and Excel for data analysis.
Week 2	Days 7-12	Prepare and query: data cleaning, SQL queries and statistics for analytics, each with a concept session and a hands-on task.
Week 3	Days 13-18	Visualise and explore: data visualization, Power BI / Tableau basics and exploratory data analysis.
Week 4	Days 19-24	Present and apply: dashboard creation, insight reporting, a case study exercise and a mock practical exercise.
Week 5	Days 25-28	Project closure: supervisor review and feedback, internship report drafting, presentation preparation and final review.

5. Week-wise Narrative

Week 1: The first week was devoted to orientation and foundations. After company induction, the student was introduced to data analytics and studied the types of analytics, the analytics process and common tools. The week ended with a study of Excel for data analysis - formulas, sorting, filters, pivot tables and charts - supported by hands-on work on a sample sales sheet.

Week 2: In the second week the study moved to data cleaning - removing duplicates, fixing errors and handling missing values - followed by SQL queries, covering SELECT, WHERE, GROUP BY and JOIN, and statistics for analytics, covering mean, median, variance, correlation and distributions.

Week 3: The third week focused on data visualization, including choosing charts and presenting data clearly; Power BI / Tableau basics, including connecting data, building visuals and simple reports; and exploratory data analysis, including summarising, visualising and finding patterns in datasets.

Week 4: In the fourth week the student studied dashboard creation - designing an interactive dashboard with KPIs and filters - and insight reporting - turning analysis into findings and recommendations. A case study exercise and a mock practical exercise were completed in this week.

Week 5: The last week was devoted to project closure: supervisor review and feedback, drafting of the internship report, preparation of a presentation and final review before submission.

6. Topic-wise Learning in Detail

6.1 Data Analytics Overview

Concept: Types of analytics, the analytics process and common tools. **Practice carried out:** Classifying sample business questions as descriptive, diagnostic, predictive or prescriptive, writing the analytics process for a sample shop and listing the tool used at each step.

6.2 Excel for Data Analysis

Concept: Formulas, sorting, filters, pivot tables and charts. **Practice carried out:** Applying SUM, AVERAGE, COUNTIF, SUMIF, IF and lookup formulas to a sample sales sheet, sorting and filtering records, building a pivot table by category and month and inserting charts.

6.3 Data Cleaning

Concept: Removing duplicates, fixing errors and handling missing values. **Practice carried out:** Cleaning a messy dummy sales file: removing duplicates, trimming spaces, standardising city names and dates, handling missing values and recording every change in a cleaning log.

6.4 SQL Queries

Concept: SELECT, WHERE, GROUP BY, JOIN and basic queries. **Practice carried out:** Writing SQL queries on a sample sales database: SELECT with WHERE and ORDER BY, GROUP BY with aggregate functions, HAVING, and a JOIN between sales and customers tables.

6.5 Statistics for Analytics

Concept: Mean, median, variance, correlation and distributions. **Practice carried out:** Calculating mean, median, mode, variance and standard deviation of daily sales in Excel, finding the correlation between footfall and sales and describing the shape of a distribution.

6.6 Data Visualization

Concept: Choosing charts, building graphs and presenting data clearly. **Practice carried out:** Choosing chart types for ten sample questions, building bar, line, pie and scatter charts in Excel and redesigning a cluttered chart into a clear one.

6.7 Power BI / Tableau Basics

Concept: Connecting data, building visuals and simple reports. **Practice carried out:** Connecting a sample data file to Power BI Desktop or Tableau Public, building a few visuals, writing simple measures and preparing a one-page report.

6.8 Exploratory Data Analysis

Concept: Summarising, visualising and finding patterns in datasets. **Practice carried out:** Summarising a sample dataset, drawing histograms and box plots, checking missing values and outliers and listing patterns and questions for further study.

6.9 Dashboard Creation and Insight Reporting

Concept: Designing an interactive dashboard with KPIs and filters; turning analysis into findings and recommendations. **Practice carried out:** Designing a dashboard with KPI cards, charts and slicers, and writing a one-page insight report with findings, evidence and recommendations.

7. Rules, Ethics and Good Practice - Points Studied

Data analytics operates within a legal and ethical framework. Personal data, such as a name, phone number or purchase history that can identify a person, may be collected and used only with the person's consent and for a stated purpose under the Digital Personal Data Protection Act, 2023. The Information Technology Act, 2000 governs electronic records and prohibits unauthorised access to computer systems and data. Datasets, reports, dashboards and software are protected by the Copyright Act, 1957, and the licence terms of data sources and tools must be followed. Employers and clients also set confidentiality and data-security rules, which must be respected even after the work is over.

Good practice includes working on a copy and never on the only original file, keeping a cleaning log, removing or masking personal identifiers before sharing, protecting files with passwords and limited access, not uploading confidential data to public websites or public dashboards, checking totals against the source, stating assumptions and limitations, avoiding misleading charts, giving credit for data and ideas taken from others, and reporting errors found in one's own earlier work honestly.

8. Process Flows Studied

Process mapping was used to understand how an analysis moves from a question to a decision and who is responsible at each step. The main flows studied were:

- **Analytics project:** Business question > Data collection > Data cleaning > Analysis > Visualisation > Insight and recommendation > Decision and follow-up.
- **Excel analysis:** Import data > Convert to table > Clean > Add formulas > Pivot table > Charts > Check totals > Save and document.
- **Data cleaning:** Keep raw copy > Profile data (counts, blanks, types) > Remove duplicates > Fix errors and formats > Handle missing values > Check outliers > Validate totals > Write cleaning log.
- **SQL query:** Understand the question > Identify tables and columns > Write SELECT with filter > Group and aggregate > Join tables if needed > Sort > Check result against known totals.
- **Dashboard:** Audience and purpose > Choose and define KPIs > Prepare data model > Build visuals > Add filters > Test with users > Publish and refresh.
- **Insight reporting:** Finding > Evidence (chart or number) > Meaning > Recommendation > Priority and owner > Follow-up.

9. Worked Examples with Sample (Dummy) Data

The examples below use a fictitious local business, *Sunrise Mart*, a general store selling grocery, snacks, personal care and stationery items in a town in Bihar. All names, numbers and results are dummy figures for learning.

Example A: Business question and type of analytics

The owner of Sunrise Mart asked four questions, which show the four types of analytics. "How much did we sell last month, and in which category?" is **descriptive**. "Why did sales fall in April?" is **diagnostic** - the student compared April with March by category and by weekday, and checked stock-outs. "How much will we sell in July?" is **predictive** - a simple trend based on the last six months was used, with the warning that it is only an estimate. "What stock should we order, and when?" is **prescriptive** - recommendations were made on the basis of the earlier findings. The student wrote each question, the data needed, the tool to be used and the expected output in a one-page analysis plan before starting any work.

Example B: Excel summary and pivot table

A sample week of sales was entered as a table with the columns Product, Category, Quantity, Price and Amount, where Amount = Quantity x Price. A pivot table with Category in rows and the sum of Amount in values gave the summary below.

Category	Total Amount (Rs.)	% of total sales
Grocery	12,300	47.9%
Personal Care	5,600	21.8%
Stationery	4,000	15.6%
Snacks	3,800	14.8%
Grand total	**25,700**	**100%**

The grand total of the pivot table (Rs. 25,700) was checked against the sum of the Amount column in the raw data and matched. The percentages are rounded to one decimal place, so they may not add up to exactly 100. The student noted that Grocery alone gave nearly half of the sales, and used SUMIF, for example =SUMIF(B2:B9,"Grocery",E2:E9), to confirm the Grocery figure independently.

Example C: Data cleaning log

A dummy file of Sunrise Mart sales had 520 records. The cleaning steps and their effect were recorded in a log, as shown below.

Step	Problem found	Action taken	Result
1	Repeated records of the same order	Removed duplicates using order ID and product	520 records reduced to 500 (20 removed)
2	Extra spaces and different spellings: "Gaya", "gaya " and "GAYA"	TRIM and PROPER, then Find and Replace	41 city entries standardised to "Gaya"
3	Dates typed as text in different formats	Converted to a single date format	8 invalid dates corrected after checking the register
4	Missing price in 12 records	Filled with the median price of the same product; flagged in a new column	0 blanks left in price; 12 flagged (2.4% of 500)
5	Validation	Compared record counts and total quantity before and after	Only the 20 duplicate records explained the difference

Every change was written in the log so that the supervisor could check it, and the raw file was kept unchanged in a separate sheet.

Example D: SQL queries

A sample database had a table *sales* (order_id, product, category, quantity, amount, customer_id) and a table *customers* (customer_id, name, city). The following queries were written. The first gives total sales for each category, using the same sample week as Example B.

```
SELECT category, SUM(amount) AS total_sales
FROM sales
GROUP BY category
ORDER BY total_sales DESC;
```

The result was Grocery 12,300; Personal Care 5,600; Stationery 4,000; Snacks 3,800, matching the pivot table. To list only the categories with more than Rs. 5,000 of sales, HAVING was added after GROUP BY:

```
SELECT category, SUM(amount) AS total_sales
FROM sales
GROUP BY category
HAVING SUM(amount) > 5000;
```

This returned only Grocery (12,300) and Personal Care (5,600). To find sales by city, the two tables were joined on customer_id:

```
SELECT c.city, SUM(s.amount) AS total_sales
FROM sales s
INNER JOIN customers c ON s.customer_id = c.customer_id
GROUP BY c.city
ORDER BY total_sales DESC;
```

The result was Gaya 15,200; Patna 6,500; Nawada 4,000, whose total of Rs. 25,700 agreed with the grand total. The student learned that a result must always be compared with a known total, because a wrong join can silently repeat or drop rows.

Example E: Statistics on daily sales

Sales (in Rs.) and the number of customers entering the shop (footfall) for one week were:

Day	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Sales (Rs.)	4,000	4,500	3,800	5,200	4,200	6,100	4,200
Footfall	82	88	70	96	90	118	74

Mean sales = 32,000 / 7 = Rs. 4571.43. After sorting (3,800; 4,000; 4,200; 4,200; 4,500; 5,200; 6,100) the median is Rs. 4,200 and the mode is also Rs. 4,200. The median is lower than the mean because the high Saturday value pulls the mean upward. The sample standard deviation (STDEV.S) is about Rs. 809.76, which shows that sales vary considerably from day to day. The correlation between footfall and sales (CORREL) is r = 0.94, a very strong positive relationship, which means that days with more customers had higher sales. The student noted that seven days are too few to be certain and that correlation does not prove that footfall causes sales.

Example F: Choosing the right chart

Question	Chart chosen	Reason
Which category sells the most?	Bar chart (sorted)	Easy comparison of amounts across categories
How did sales change month by month?	Line chart	Shows the trend over time
What share of sales does each of four categories have?	Pie / donut or stacked bar	Part-to-whole with only a few parts
Do busier days have higher sales?	Scatter plot	Shows the relationship between two numbers
How are individual order values spread?	Histogram	Shows the shape of the distribution
Which city and category combination is strongest?	Heat map / matrix	Two-way comparison with colour intensity

A cluttered chart with a three-dimensional pie of twelve slices was redesigned as a sorted bar chart with a clear title and direct labels, which made the message readable in a few seconds.

Example G: Power BI measures and relationship

The Sales and Customers tables were loaded into Power BI Desktop and connected by customer_id (one customer to many sales). The following measures were written in DAX:

```
Total Sales = SUM(Sales[Amount])
Total Orders = DISTINCTCOUNT(Sales[OrderID])
Average Order Value = DIVIDE([Total Sales], [Total Orders])
Profit Margin % = DIVIDE([Total Profit], [Total Sales])
```

DIVIDE was used instead of the division sign because it handles a zero denominator safely. In Tableau, the equivalent calculated field for average order value is SUM([Amount]) / COUNTD([Order ID]). The student observed that measures defined once in the model give the same figure on every page, which avoids the confusion of two different totals.

Example H: Exploratory analysis of monthly sales

Monthly sales of Sunrise Mart for six months (in Rs. thousand) and the change from the previous month are shown below.

Month	Sales (Rs. thousand)	Change vs previous month
Jan	380	-
Feb	410	+7.9%
Mar	440	+7.3%
Apr	400	-9.1%
May	490	+22.5%
Jun	540	+10.2%

Total sales = Rs. 2,660 thousand and the monthly average = 2,660 / 6 = Rs. 443.3 thousand. Growth from January to June = (540 - 380) / 380 x 100 = 42.1%. The exploration showed a steady rise except for April, which fell by 9.1% from March. The student listed follow-up questions: was there a stock-out, a holiday or a competitor offer in April? The best month, June, was also checked by category to see which products drove the increase.

Example I: Dashboard KPIs

A one-page dashboard was designed for the owner with KPI cards at the top, a monthly trend line in the middle, a category bar chart and a city-wise table below, and slicers for month, city and category. The sample KPIs for the six months were:

KPI	Definition	Sample value
Total sales	Sum of the amount of all orders	Rs. 26,60,000
Total orders	Number of distinct orders	5,320
Average order value	Total sales / total orders = 2,660,000 / 5,320	Rs. 500
Profit margin	Profit / total sales x 100	18% (profit Rs. 4,78,800)
Growth	(June - January) / January x 100	+42.1%

Each KPI card was shown with a comparison to the previous period, the definitions were written in a notes panel, and the dashboard was tested by checking that every number matched the Excel summary.

Example J: Insight report extract

Finding	Evidence	Recommendation
Grocery is the biggest category	Rs. 12,300 of Rs. 25,700 in the sample week (47.9%)	Never allow stock-outs; ask suppliers for bulk rates
Saturday is the strongest day	Rs. 6,100 against a weekly average of Rs. 4571.43	Add staff and stock more on weekends
Sales dipped in April	-9.1% compared with March	Check stock and offers of that month; plan festival offers early
Footfall and sales move together	Correlation r = 0.94 in the sample week	Run local WhatsApp and poster offers to bring more customers

The report began with a three-line summary, listed these four findings with one chart each, stated that the data covered only a sample week and six months of dummy figures, and ended with the next steps. The student also learned three simple formulas used in reports: percentage change = (new - old) / old x 100; average order value = total sales / number of orders; and profit margin = profit / sales x 100.

10. Project: Business Data Analysis & Dashboard Project

Scope. The internship work was organised around a project in which a sample business dataset was cleaned, analysed with Excel and SQL, explored with statistics and charts, presented in a Power BI / Tableau dashboard and summarised in an insight report for a fictitious business, using sample datasets and dummy data.

Steps followed. (i) Studied the analytics types and process and wrote the business questions; (ii) cleaned the raw sample file, removed duplicates, fixed errors, handled missing values and prepared a cleaning log; (iii) summarised the data in Excel with formulas, pivot tables and charts; (iv) wrote SQL queries with filters, grouping and joins and compared the results with the Excel totals; (v) calculated descriptive statistics and correlation and explored the dataset for patterns and outliers; (vi) designed charts, built a dashboard with KPIs and filters in Power BI or Tableau, wrote the insight report and reviewed the work with the supervisor.

Skills and tools. MS Excel, SQL, Power BI / Tableau, Statistics, Data Cleaning and Data Visualization.

Outputs. Analysis plan, cleaning log and cleaned dataset, Excel workbook with pivot tables and charts, SQL query sheet, statistics summary, EDA notes, dashboard file, insight report, and this internship report.

11. Case Study Exercise: A Shop with Data but No Decisions

A case study exercise was completed on a typical data analytics problem. The business and figures are illustrative and not taken from any real organisation.

Background. A medium-sized general store (fictitious) had recorded two years of sales in notebooks and in several Excel files made by different staff members. The owner paid a freelancer to build a dashboard, expecting it to show how to increase profit.

Problem observed. The combined file had about 18,000 rows. About 6% of the rows were duplicates, so sales were overstated, and the 120 products appeared under about 340 different spellings, so product totals were wrong. Dates were typed as text in three formats, and many prices were blank. The dashboard had 25 charts on one page, including a three-dimensional pie chart, no definition of "profit" or "growth", and no filters. The figures on the dashboard did not match the owner's own Excel totals, so the owner stopped trusting it and went on taking decisions by habit. Nobody had written down what question the dashboard was supposed to answer.

Analysis. The main problem was not the lack of data or of tools but the lack of data quality, clear definitions and a purpose. Duplicates and spelling variations produced wrong numbers, and a wrong number shown in a beautiful chart is more dangerous than no chart. A page with too many visuals hid the important message, and without agreed KPI definitions every person calculated profit differently. Because no one had checked the dashboard against the source, the errors were discovered only by the owner.

Recommendations. (1) Begin with two or three business questions, such as which products and days give most profit. (2) Clean the data first: remove duplicates, create a product master list and use it to standardise names, convert dates and fill or flag missing prices, keeping a cleaning log. (3) Store the cleaned data in one table or small database and enter new data through a form with validation. (4) Define each KPI in writing. (5) Rebuild the dashboard with five to seven visuals, KPI cards with comparison to the previous period and slicers for month, category and product. (6) Check every figure against the source before the owner sees it. (7) Write a monthly one-page insight report with findings and actions, and review the actions in the next month.

Expected result and lesson learned. With clean data and clear definitions, the owner can trust the numbers and act on them, for example by stocking fast-moving items before weekends. The lesson is that analytics is a chain: question, data, cleaning, analysis, visual and action must all be sound, and the weakest link decides the quality of the decision.

12. Mock Practical Exercise: An End-to-End Mini Analysis

In the mock practical exercise the student carried out a complete mini analysis of a sample (dummy) sales dataset of a local business, from the first question to the final report, using the tools and methods practised earlier. The steps followed were:

1. Question: wrote the business question, the audience, the data needed and the expected output.

2. Data preparation: kept a raw copy, profiled the data, removed duplicates, standardised text and dates, handled missing values and wrote the cleaning log.
3. Excel: built formulas, a pivot table by category and month and two charts, and checked the totals.
4. SQL: wrote queries for total sales by category, by month and by city, with a HAVING condition and a JOIN, and compared the results with Excel.
5. Statistics: calculated mean, median, standard deviation and correlation and checked for outliers.
6. Visualisation: chose suitable charts for each message and removed clutter.
7. Dashboard: built a one-page dashboard with KPI cards and slicers in Power BI or Tableau and tested the filters.
8. Reporting: wrote a one-page insight report with three findings, evidence, recommendations and limitations.

The exercise showed that every step depends on the one before it: a vague question leads to unfocused work, dirty data leads to wrong results, a wrong join spoils the totals, a poor chart hides the message and an honest report is the base of the next decision.

13. Sample Checklists Prepared

Checklists were prepared as part of the hands-on tasks. They reduce errors and make routine work easier to review.

Excel and Data Cleaning Checklist

- Raw data copied and kept unchanged in a separate sheet.
- One header row, one record per row, no merged cells in the data.
- Duplicates, blanks, wrong data types and impossible values checked.
- Text and dates standardised; spellings made consistent.
- Missing values handled by a stated rule and flagged.
- Formulas used instead of typed totals; absolute references checked.
- Totals and record counts compared before and after cleaning; cleaning log written.

SQL Checklist

- Business question and the required tables and columns identified.
- Column names written instead of SELECT *; table aliases used.
- WHERE used for row conditions and HAVING for group conditions.
- NULL values tested with IS NULL.
- Join keys checked; join type chosen deliberately.
- Result compared with a known total or the Excel summary.
- Only read queries run on practice data; any UPDATE or DELETE avoided.

Statistics and EDA Checklist

- Size, period and column meanings of the dataset noted.
- Mean, median, minimum, maximum and spread calculated for numeric columns.
- Sample (S) or population (P) function chosen correctly.
- Histograms or box plots drawn; outliers checked, not deleted blindly.
- Correlation interpreted with the warning that it does not prove cause.
- Number of records behind every percentage noted.
- Patterns, surprises and further questions written down.

Visualization and Dashboard Checklist

- Audience and purpose of the dashboard stated.
- KPIs chosen, defined in writing and compared with a target or earlier period.
- Chart type fits the question; axes labelled; bar charts start at zero.
- No more than a handful of visuals on a page; consistent colours and formats.

- Slicers and filters tested; data source and refresh date shown.
- Every figure matched with the source before sharing.
- No personal or confidential data shown or published publicly.

Insight Reporting Checklist

- Executive summary written in a few simple lines.
- Each finding supported by evidence, a chart or a number.
- Reasons or likely causes explained without technical jargon.
- Recommendations specific, prioritised and linked to an owner.
- Assumptions, limitations and data period stated honestly.
- Same method followed in each regular report.
- Next steps and follow-up date added.

14. Tools and Resources

Learning materials, MS Excel (including pivot tables and Power Query) and Google Sheets, free SQL practice tools such as SQLite with a database browser or a free MySQL or PostgreSQL set-up, Power BI Desktop, Tableau Public (with dummy or open data only), MS Word and PowerPoint for documentation and presentation, Microsoft Learn and Tableau help pages for reference, assignments and mentor feedback were used during the internship.

15. Programme at a Glance

The programme was in the domain of Data Analytics with RKNEXORA PRIVATE LIMITED as the agency. It required a minimum of 120 hours over 28 learning days, organised in three phases: Orientation (Days 1-2), Learning and Practice (Days 3-22) and Project and Review (Days 23-28). Attendance was recorded in the Internship Attendance Log, signed by the supervisor (see Attendance Sheet).

LEARNING OUTCOME

a. Knowledge Acquired

1. The types of analytics (descriptive, diagnostic, predictive and prescriptive), the steps of the analytics process and the common tools.
2. How to use Excel formulas, sorting, filters, pivot tables and charts to summarise data.
3. How to find and correct duplicates, spelling variations, wrong formats and missing values, and how to keep a cleaning log.
4. How to write SQL queries with SELECT, WHERE, GROUP BY, HAVING and JOIN and check them against known totals.
5. How to calculate and interpret mean, median, mode, variance, standard deviation, correlation and distributions.
6. How to choose honest charts and build visuals, measures and reports in Power BI or Tableau.
7. How to explore a dataset, design a KPI dashboard and write an insight report with findings and recommendations.

Glossary of Terms Learned

- **Data analytics** - Collecting, cleaning, examining and interpreting data to support decisions.
- **Descriptive / Diagnostic / Predictive / Prescriptive** - What happened / why it happened / what may happen / what to do.
- **Dataset / Record / Field** - A collection of data / one row / one column of a dataset.
- **Data cleaning** - Finding and correcting duplicates, errors and missing values before analysis.
- **Missing value / Outlier** - A blank entry / a value far from the others that needs checking.
- **Pivot table** - An Excel tool that summarises a long list by categories.
- **Primary key / Foreign key** - A unique column in a table / a column that links to another table.
- **SQL** - Structured Query Language, used to ask questions of a database.
- **JOIN** - A SQL operation that combines rows of two tables on a matching column.
- **Aggregate function** - SUM, COUNT, AVG, MIN or MAX: one value calculated from many rows.
- **GROUP BY / HAVING** - Groups rows for aggregation / filters the groups afterwards.
- **Mean / Median / Mode** - Average / middle value / most frequent value.
- **Variance / Standard deviation** - Measures of how widely values are spread around the mean.
- **Correlation** - Strength of the linear relationship between two numbers, from -1 to +1.
- **Distribution** - The way values are spread, for example normal or skewed.
- **EDA** - Exploratory data analysis: a first careful look at a dataset to find patterns.
- **Data visualization** - Presenting data as charts and graphs.
- **KPI** - Key performance indicator: a defined measure of success.
- **Dashboard** - A one-screen, interactive display of key measures.
- **Power Query / DAX** - Data preparation tool / formula language of Power BI.
- **Slicer / Filter** - A control that limits the data shown in a report.
- **Insight** - A finding supported by evidence that suggests an action.

b. Skills Learned

Technical skills. Using Excel formulas and pivot tables; cleaning and validating data; writing SQL queries with filters, grouping and joins; calculating descriptive statistics and correlation; choosing and designing charts; connecting data and building measures and visuals in Power BI or Tableau; exploring datasets; designing KPI dashboards with filters; and writing insight reports.

Professional skills. Analytical and numerical thinking; logical reasoning; attention to detail and checking of results; clear writing and presentation; business understanding; time management; ethical and legal awareness; teamwork and acceptance of feedback; and documentation, report writing and presentation.

These skills were developed through repeated practice and daily documentation. Cleaning and checking totals built accuracy, SQL and statistics improved logical thinking, designing charts and dashboards improved clarity and creativity, writing insight reports improved communication, and keeping a cleaning log built a habit of recording every step.

c. Observed Attitude and Gained Values

1. Accuracy first: a beautiful chart built on wrong data is worse than no chart.
2. Honesty and transparency: results must be reported truthfully, assumptions stated and mistakes shown.
3. Respect for privacy and confidentiality: data about people and businesses must be protected.
4. Curiosity and learning: tools and methods change, and an analyst must keep learning.
5. Discipline and punctuality: regular reporting at fixed timings and daily attendance records.

d. Challenges Faced

Understanding ideas such as JOIN types, standard deviation, correlation and DAX measures was handled by using a glossary, small worked examples in Excel and mentor guidance.

Learning several tools (Excel, SQL and Power BI or Tableau) in a short time was handled by practising each tool on a small task the same day and keeping short step notes for revision.

Choosing the right chart and the right KPIs was difficult at first; this was handled by starting from the business question, studying good and bad examples and checking the choices with the supervisor.

Working without real business data was handled by using sample datasets, dummy figures and clear assumptions written beside each calculation.

Maintaining consistent documentation was handled by using daily log entries and reviewing feedback to improve documentation progressively.

Managing time between practice, documentation and report writing was handled by dividing work week by week and using review days for corrections.

e. Career Pathways

The knowledge gained supports careers such as Data Analyst, Business Analyst, MIS Executive, Reporting Analyst, Business Intelligence (BI) Developer, Research Analyst, Operations Analyst, Junior Data Engineer and, after further study, Data Scientist. Opportunities exist in banks and financial companies, e-commerce and retail, healthcare, education, government and public-sector offices, consulting firms, start-ups and local businesses, and as a freelancer or consultant. Related certifications and study options include the Microsoft Power BI Data Analyst certification, Tableau certifications, the Google Data Analytics Professional Certificate, courses in SQL and Excel, BCA or MCA, MBA in Business Analytics, and M.Sc in Statistics or Data Science. Because tools change quickly, continuous learning and a portfolio of real projects are essential.

CONCLUSION

The internship in Data Analytics was a valuable learning experience that integrated academic knowledge with practical activity. The programme achieved its objectives through structured learning, guided practice, assignments, supervisor review and documentation.

The student gained a clear understanding of how data is prepared, analysed and presented - through the analytics process, Excel, data cleaning, SQL, statistics, data visualization, Power BI / Tableau, exploratory data analysis, dashboard creation and insight reporting - and acquired practical familiarity with cleaning logs, pivot tables, SQL queries, statistical summaries, charts, KPI dashboards and reports. The experience has strengthened the student's employability and provides a foundation for further study and work in analytics, business and technology.

Reflection

Looking back, the most useful lesson of the internship was that data analytics is a chain, and a chain is only as strong as its weakest link. A clear question guides the work, clean data protects the result, a correct query or formula gives the numbers, the right chart shows the message, a well-defined dashboard keeps everyone looking at the same facts and an honest report shows what to do next. The case study showed how a good-looking dashboard can mislead when duplicates and unclear definitions are ignored. Moving from concept to practice to documentation made abstract ideas concrete, and supervisor feedback showed where understanding was weak. The habit of recording each day's work also made it easier to see progress over the weeks.

Objectives and Achievement

The objective of understanding data analytics was achieved through concept sessions and an analysis plan. The objective of learning Excel, data cleaning and SQL was achieved through a pivot table summary, a cleaning log and a sheet of tested queries. The objective of learning statistics and data visualization was achieved through a statistics summary and a set of well-chosen charts. The objective of learning Power BI / Tableau, exploratory analysis, dashboard creation and insight reporting was achieved through a data model with measures, EDA notes, a KPI dashboard and an insight report. The objective of developing professional discipline was achieved through daily attendance, supervisor review and report writing.

Future Scope

The knowledge gained can support further study of analytics, statistics, business and computing, and careers in banks, retail, e-commerce, healthcare, education, government and start-ups. With growing use of digital payments, online services and data-driven governance, and with the growing use of artificial intelligence tools that still depend on clean and well-understood data, trained and ethical data analysts are increasingly needed, including in Bihar.

Suggestions for Improvement of the Programme

1. Include live practice on larger sample datasets and talks by data professionals and local business owners, with appropriate permissions.
2. Continue regular mentor feedback and progress reviews.
3. Provide more project-based activities such as analysing the anonymised records of a local business under supervision.
4. Encourage students to maintain detailed daily documentation.

References

1. Kolb, D. A. (1984). Experiential Learning: Experience as the Source of Learning and Development. Prentice Hall.
2. University Grants Commission (UGC). Internship guidelines for higher education institutions.
3. Ministry of Education, Government of India. National Education Policy 2020.
4. Microsoft. Microsoft Excel and Power BI documentation and Microsoft Learn training modules.
5. Tableau. Tableau Help and Tableau Public learning resources.
6. Official documentation of SQL databases such as MySQL, PostgreSQL and SQLite.

7. Knaflic, C. N. (2015). Storytelling with Data: A Data Visualization Guide for Business Professionals. Wiley.
8. Few, S. (2012). Show Me the Numbers: Designing Tables and Graphs to Enlighten. Analytics Press.
9. Provost, F. and Fawcett, T. (2013). Data Science for Business. O'Reilly Media.
10. Beaulieu, A. Learning SQL. O'Reilly Media.
11. Levine, D. M. et al. Statistics for Managers Using Microsoft Excel. Pearson.
12. Tukey, J. W. (1977). Exploratory Data Analysis. Addison-Wesley.
13. Government of India. The Information Technology Act, 2000; the Digital Personal Data Protection Act, 2023; the Copyright Act, 1957.
14. RK NEXORA internship programme handbook.

ATTENDANCE SHEET

The Internship Attendance Log issued by the IPO, with the supervisor's signature for each day, is to be attached on this page.

[Attach the signed Internship Attendance Log here]